Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Configuration and secrets before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Typed contracts and schemas

Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Requirements and service boundaries

Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

> Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Configuration supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Configuration and secrets trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Configuration** and **secret**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Configuration and secrets explanation against the supplied facts before discussing it.

Use the worked case for **Configuration** and **secret**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

> Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Configuration and secrets facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Configuration and secrets facts and keep your reasoning.

A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Configuration and secrets facts and keep your reasoning.

A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Configuration and secrets facts and keep your reasoning.

Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

Design a revealing follow-up. Apply the supplied Configuration and secrets facts and keep

> your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Configuration and secrets to Grounded Learning Service, identify a limit and choose the next useful test.

Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Configuration and secrets

> Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Explain the labelled step before checking the changed condition.

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

maximum_words=80 is public **configuration**. A separate protected store holds a placeholder credential.

AI ACADEMY / AI-6/4

Configuration and secrets

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Load setting

Read the reviewed maximum_words value of 80.



2. Load credential

The runtime obtains the token through its protected path.



3. Log safely

Record startup success without printing the token.

CHANGE ONE THING

Change maximum_words to 60.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Record startup success without printing the token.

1. Load credential

2. Log safely

3. Load setting

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Log safely Record startup success without printing the token. **Configuration** controls behaviour. A credential grants access and needs different storage and handling.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change maximum_words to 60. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. The configured answer limit becomes 60; the private token stays unchanged.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Repair a missing building block

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4".

Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
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> | |
> | |

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Configuration and secrets picture from its labels, then record a relevant limit.

01. Configuration and secrets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

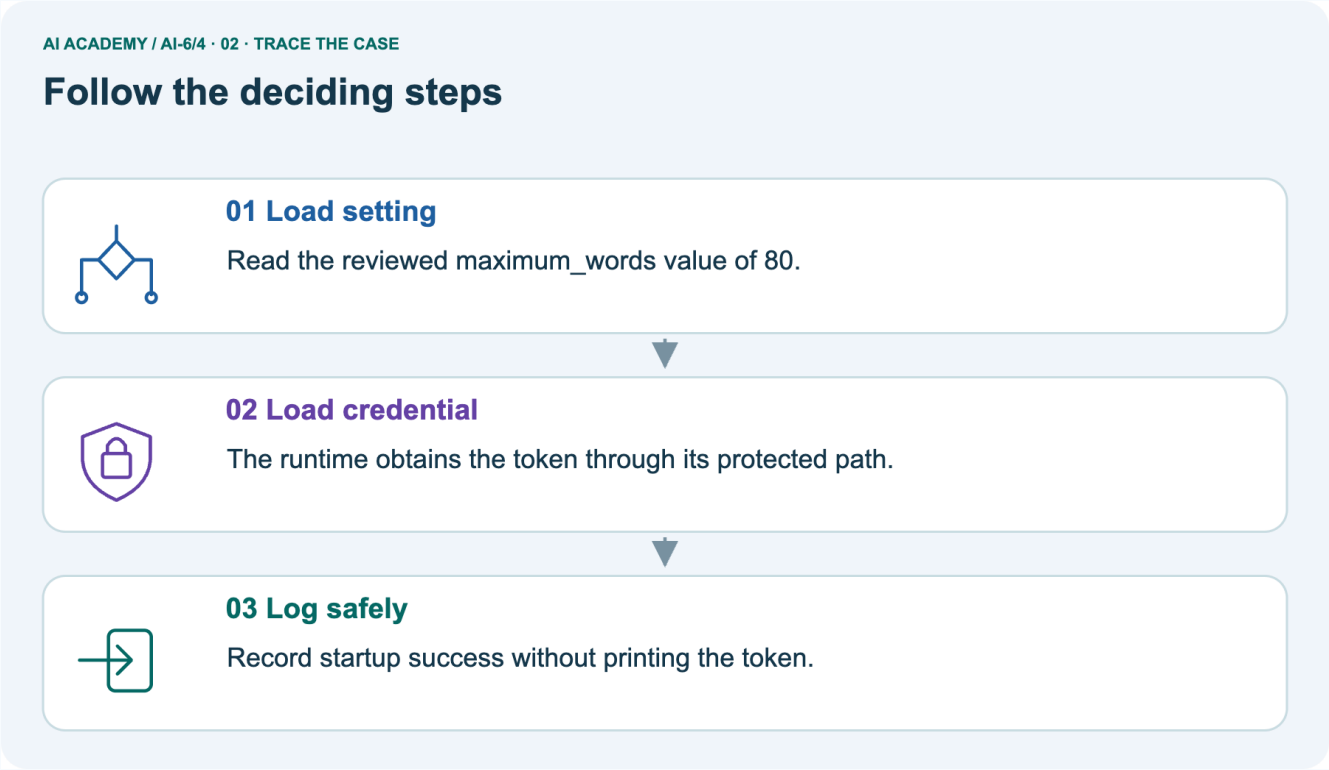
Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

A separate protected store holds a placeholder credential.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Configuration and secrets — evidence trace

AI ACADEMY / AI-6/4 · 03 · INSPECT THE LESSON MODEL

Configuration and secrets — evidence trace

Input and scope	Check or calculation	Conclusion and limit
A startup script prints every environment variable while debugging a missing data version.	The toy service reads DATA_VERSION from configuration and a signing secret from server-only input. Validate required settings at startup and fail clearly if absent; do not substitute a published default secret.	It risks exposing secrets unrelated to the error. Report the missing key name without values and validate only required fields.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration and **secret**s — **evidence** trace.

A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

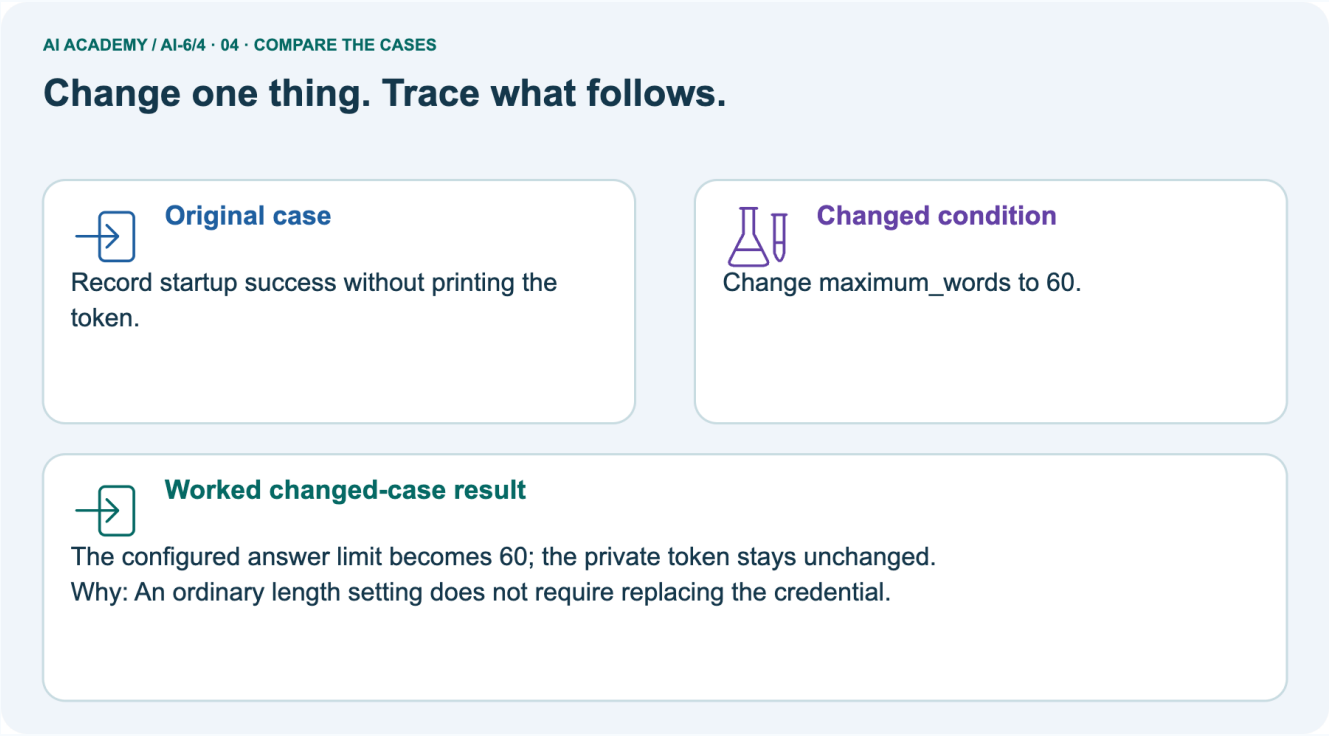
Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

Report the missing key name without values and validate only required fields.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

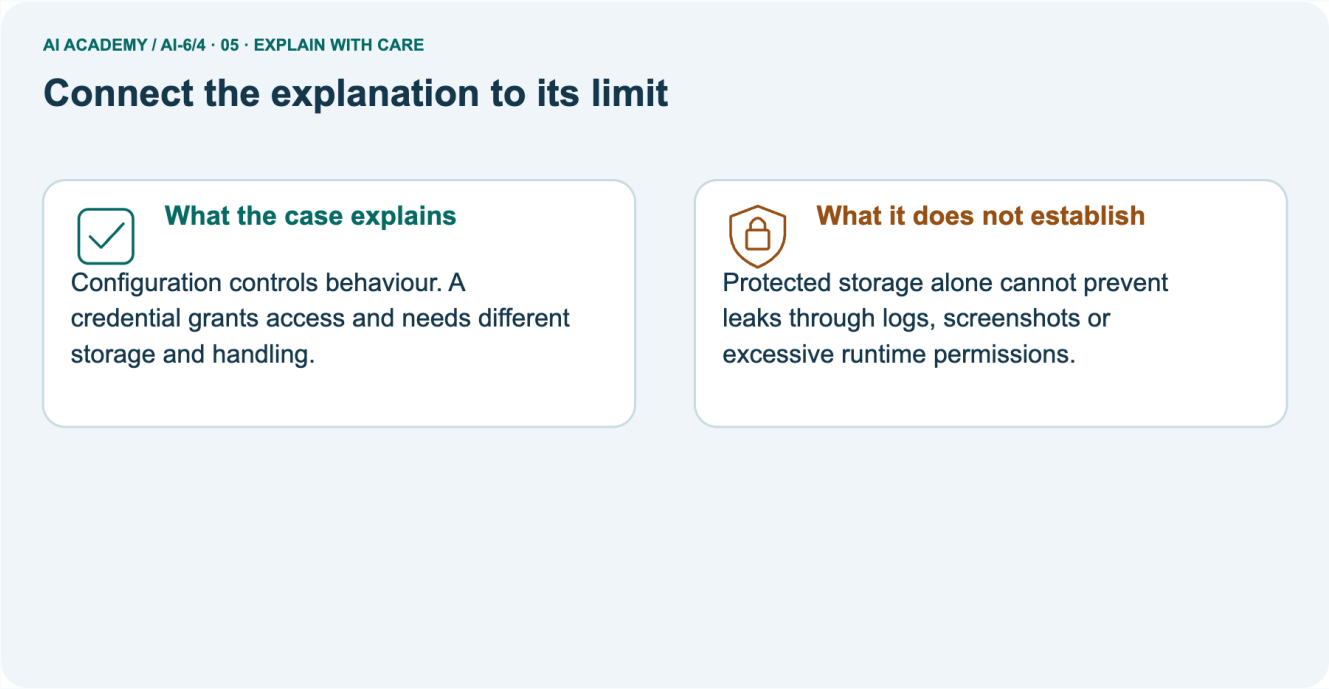
Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



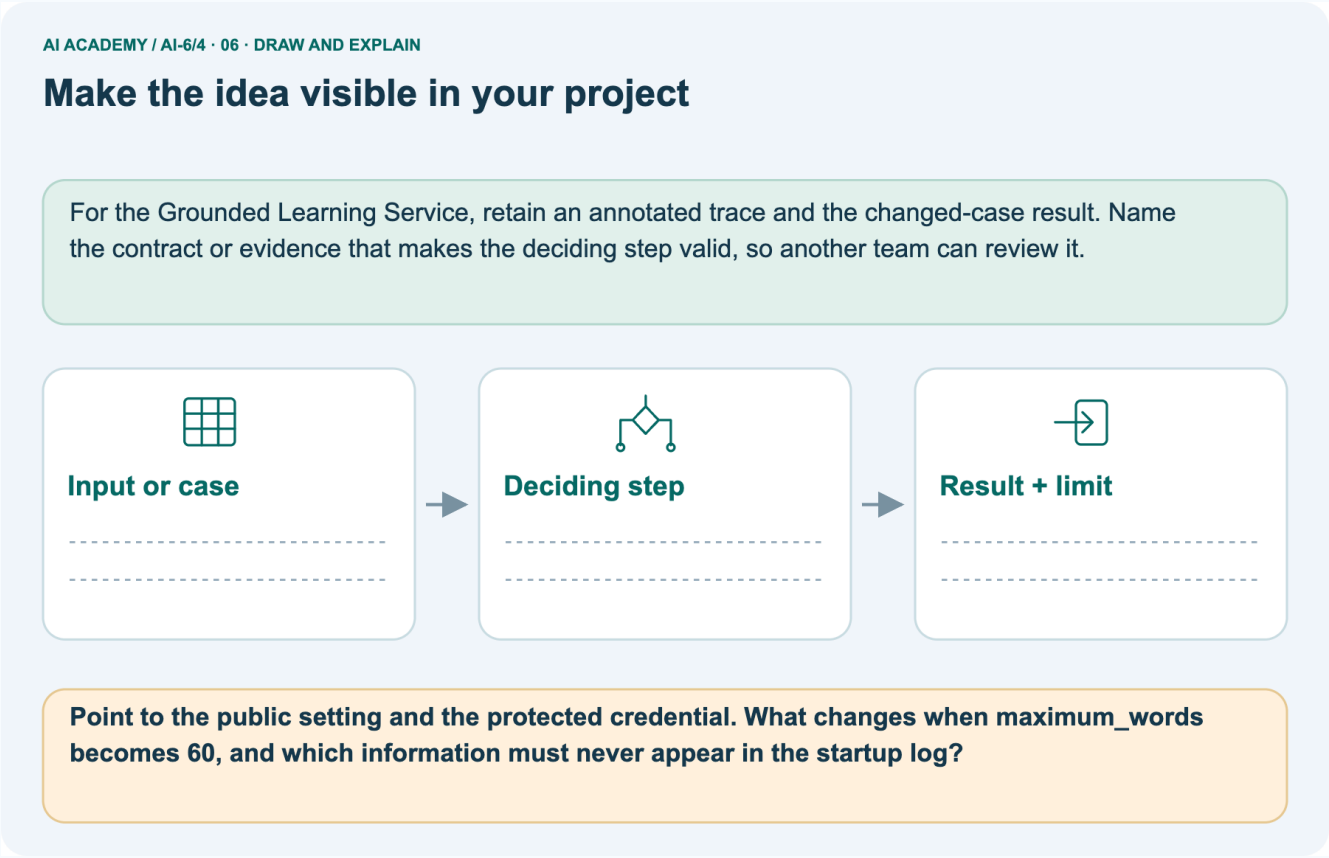
Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.